



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix



Executive Summary

- Methodologies

- Data Collection utilizing SpaceX API and Web Scraping
- Data Wrangling and Manipulation
- Exploratory Data Analysis using SQL and Visualization
- Interactive Visual Analysis utilizing Folium and Plotly Dash
- Classification modeling and Predictive Machine Learning

- Results

- Exploratory Data Analysis Insights
- Interactive Dashboard for Visual Analytics
- Predictive Analytics via Machine Learning model



Introduction

- The space industry is heating up, with SpaceX leading the way through their reusable rockets. SpaceX can recover the first stage of some of their launches, which allows them to offer commercial rocket launches for \$62 million, a great price when compared to other providers of the same service which can cost up to \$165 million per launch.
- SpaceY is an emerging competitor in the commercial rocket launch market and seeks to determine the price of each launch. Using information gathered about SpaceX, our goal is to build an interactive dashboard for our stakeholders modeled after SpaceX's launch data as well as build a machine learning model to predict if SpaceX will reuse the stage on a launch.
- This will aid SpaceY in determining the factors that influence if a rocket will land successfully, the success rate that is achievable, and the ideal operating conditions to ensure a successful landing, ultimately allowing them to compete with SpaceX.

Section 1

Methodology

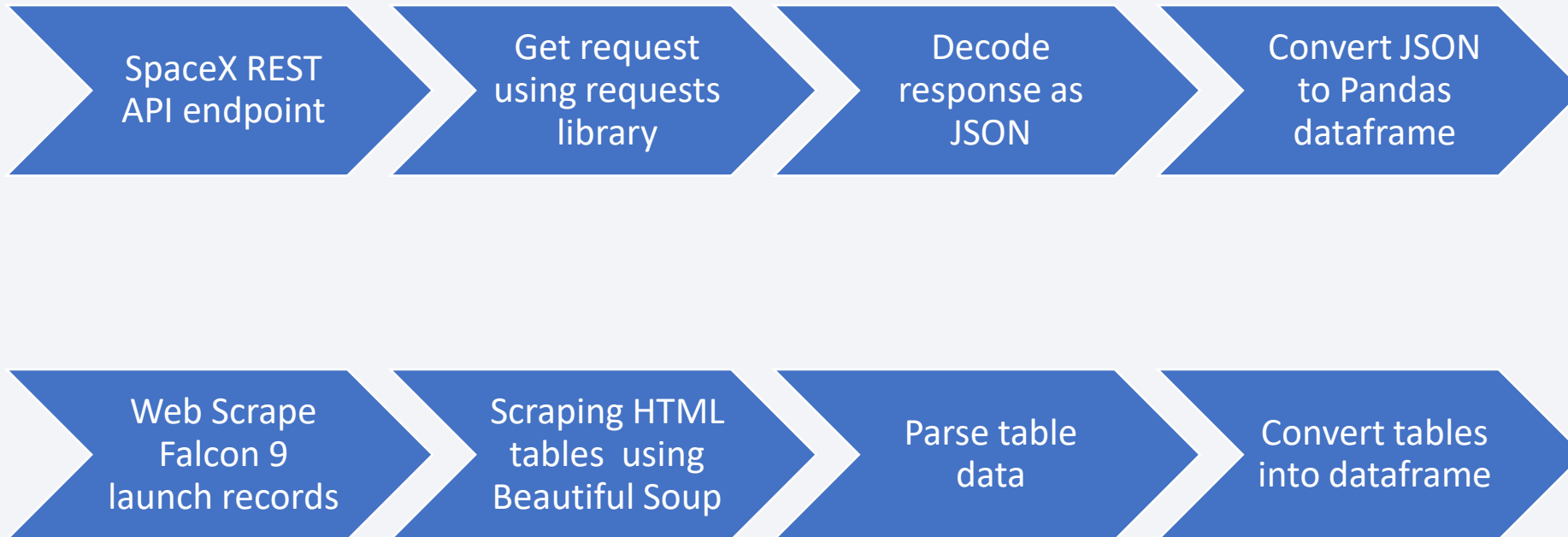
Methodology

Executive Summary

- Data collection methodology:
 - Data was gathered via SpaceX REST API and web scraping of Wikipedia launch data
- Perform data wrangling
 - Conversion of categorical features utilizing one-hot encoding to prepare them for our learning models
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Building, tuning, and evaluation of various classification models to build the most effective machine learning model

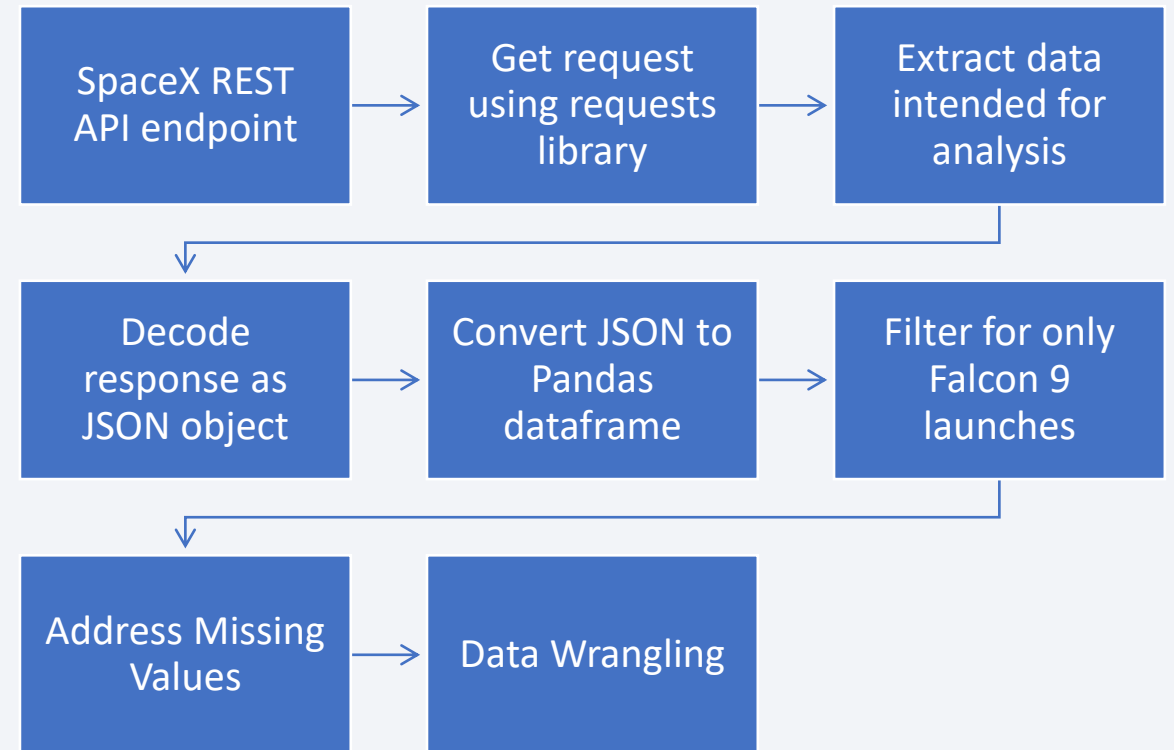
Data Collection

- All data was gathered via SpaceX REST API requests and web scraping of Wikipedia



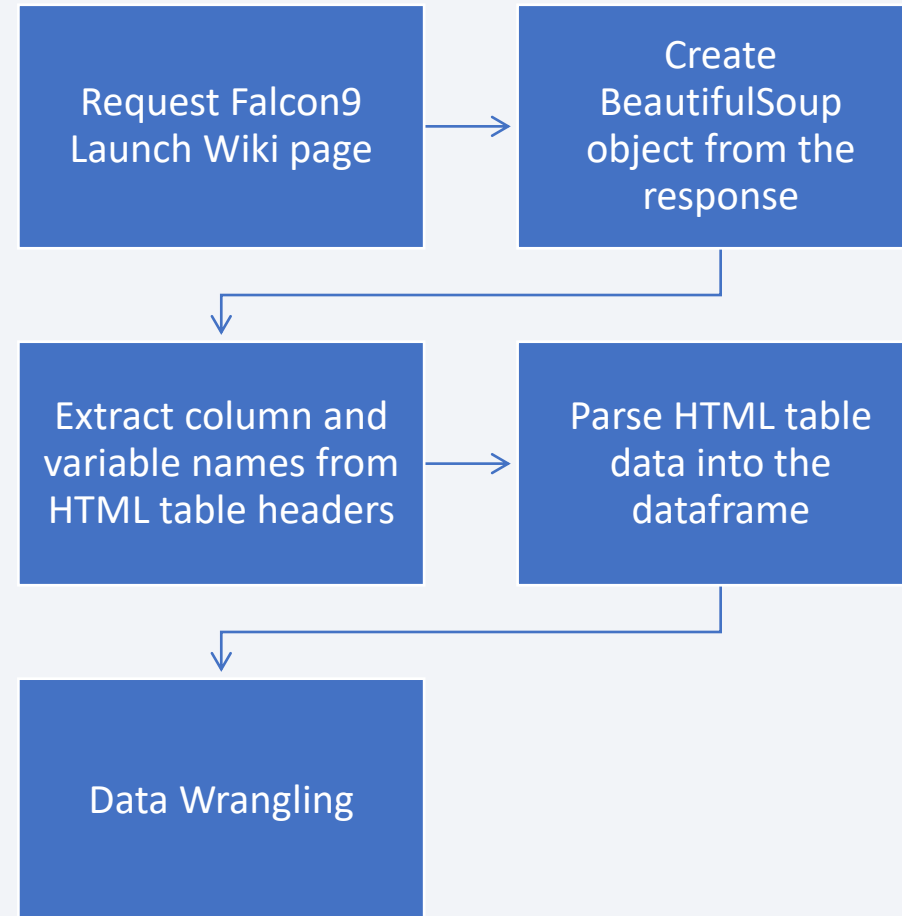
Data Collection – SpaceX API

- Using a get request on SpaceX's API, we extract the data we are interested in (booster name, launch site, payload mass, landing site, etc.) as a JSON object before converting to a dataframe to conduct cleaning and wrangling.



Data Collection - Scraping

- Using BeautifulSoup within Python allowed for web scraping of Falcon 9 launch records on Wikipedia.
- After requesting data from the URL and extracting the table we are interested in from the HTML tables, the data can be converted into a dataframe for wrangling and analysis.



Data Wrangling

- Missing values will be addressed first, followed by determining the data types of each column
- Data was processed for Exploratory Data Analysis to find trends and determine the training labels for our model.



EDA with Data Visualization

- Scatterplots were utilized for the following visuals:
 - Relationship between Flight # and Payload
 - Relationship between Flight # and Launch Site
 - Relationship between Payload and Launch Site
 - Relationship between Flight # and Orbit type
 - Relationship between Payload and Orbit type
- Other Visualization types:
 - A bar chart was utilized to visualize the success rate of each orbit type.
 - A line chart was utilized to show yearly trends of launch success

EDA with SQL

- SQL prompts:
 - Display the names of the unique launch sites in the space mission
 - Display 5 records where launch sites begin with the string 'CCA'
 - Display the total payload mass carried by boosters launched by NASA (CRS)
 - Display average payload mass carried by booster version F9 v1.1
 - List the date when the first successful landing outcome in ground pad was achieved
 - List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
 - List the total number of successful and failure mission outcomes
 - List the names of the booster versions which have carried the maximum payload mass, using a subquery
 - List the records which will display the month names, failure landing outcomes in drone ship, booster versions, and the launch site for the months in the year 2015
 - Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20

Build an Interactive Map with Folium

- Map objects created for the Folium map:
- Folium.Circle: added to highlight a specific coordinate on the map with a circle
- Folium.Marker: added to highlight a specific coordinate on the map with a marker
- MarkerCluster: simplifies a map that contains multiple markers in proximity or with the same coordinates
- Mouseposition: allows user to view coordinates of a point on the map when hovered over with the cursor
- Folium.polyline: an object used to draw a line between a launch site and its closest city, railway, and highway

Build a Dashboard with Plotly Dash

- The final dashboard allows the user to utilize a drop-down menu to filter by different launch sites
- Utilizing callback functions, when the user selects an option in the dropdown, it updates the charts within the visualization
- A Pie chart shows the ratio of successes and failures at each site
- A scatter plot shows the relationship between Outcome and Payload Mass for the various booster versions
- A range slider was added to filter by Payload

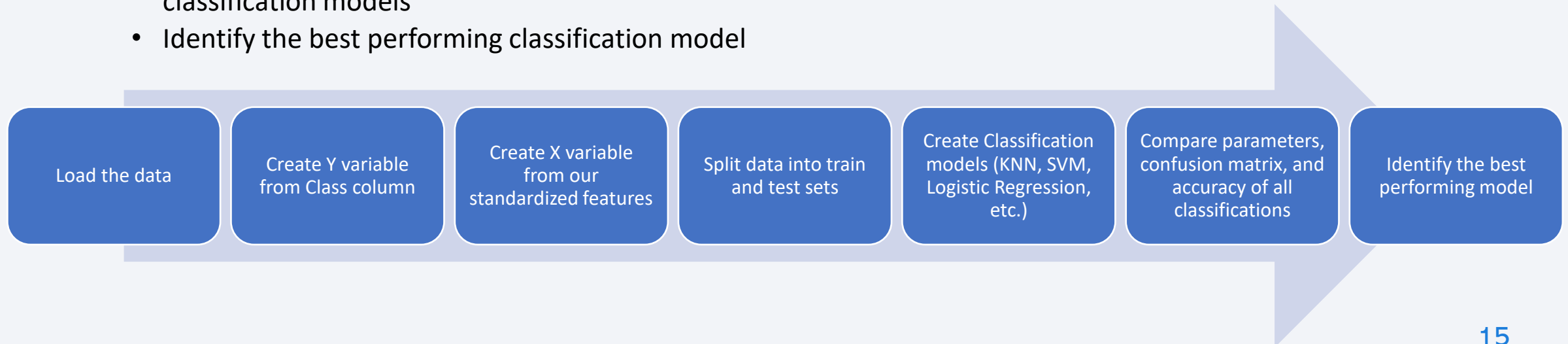
Link to the notebook hosted on Github:

https://github.com/jtnickel/Python/blob/main/IBM%20Capstone%20-%20Dash_app.py



Predictive Analysis (Classification)

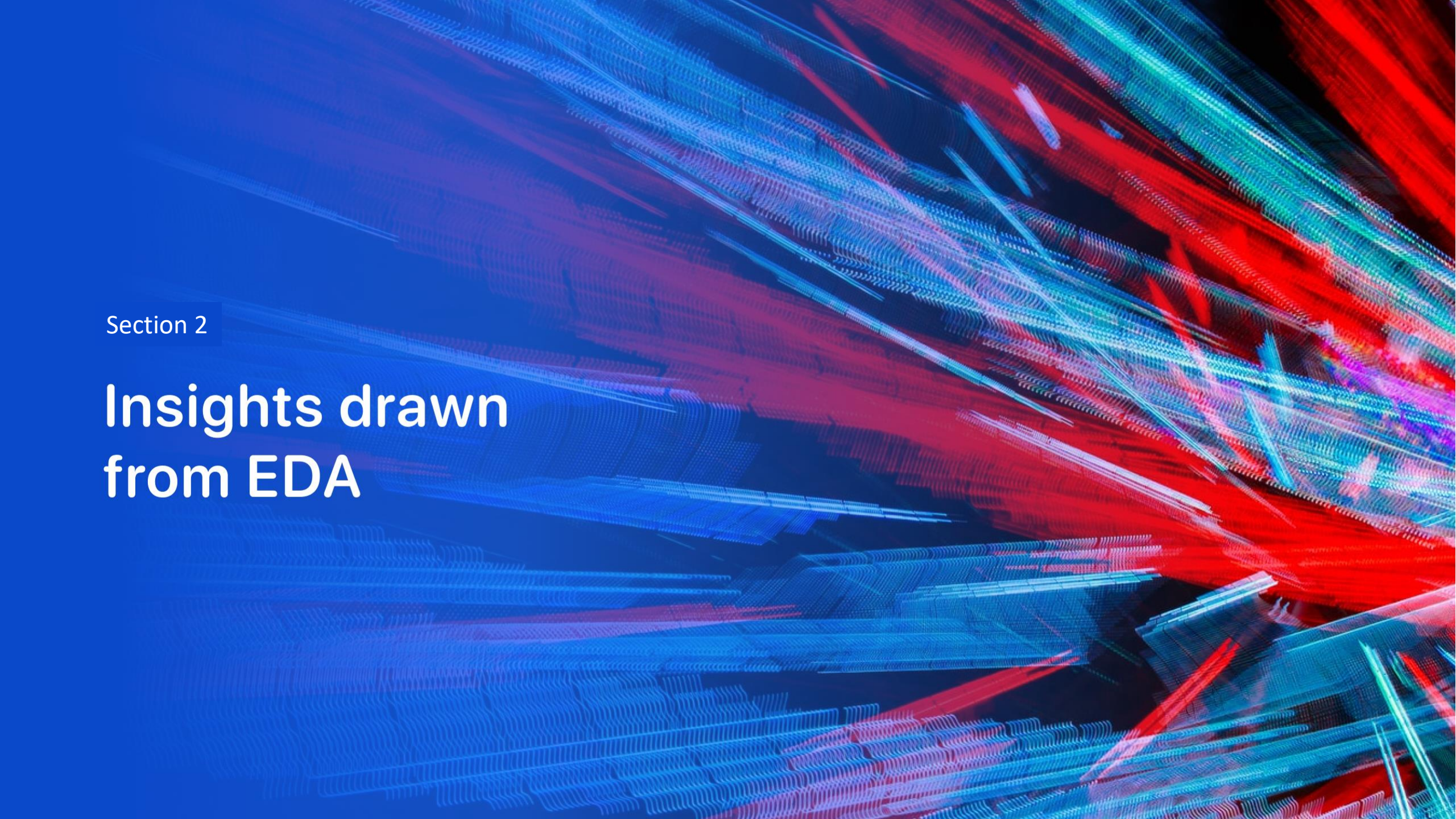
- The following outlines the process of developing and testing our models to predict whether the first stage will land or not:
 - Utilized Numpy to create an array from the Class column
 - Standardization of data
 - Split our data into train and test sets
 - Calculating the most ideal hyperparameters for KNN, SVM, Decision tree and Logistic Regression classification models
 - Identify the best performing classification model



Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results



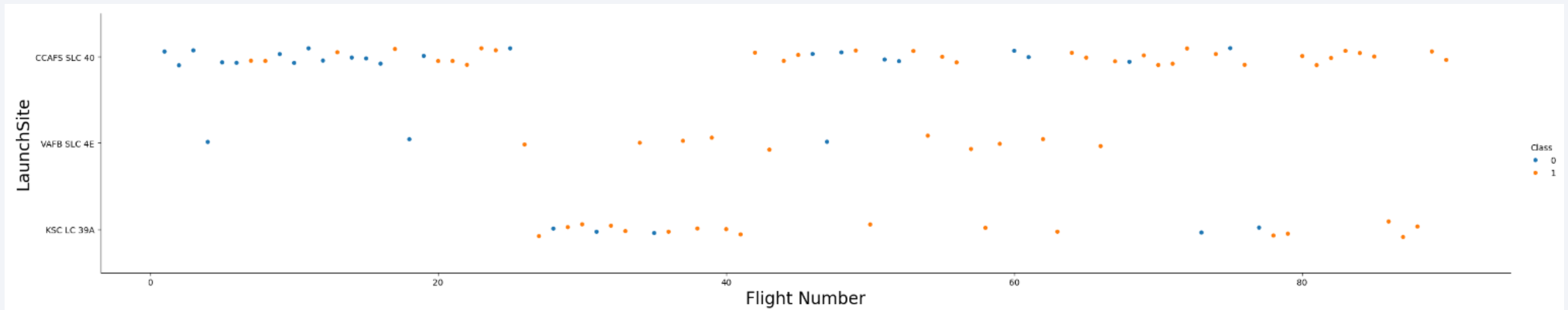
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

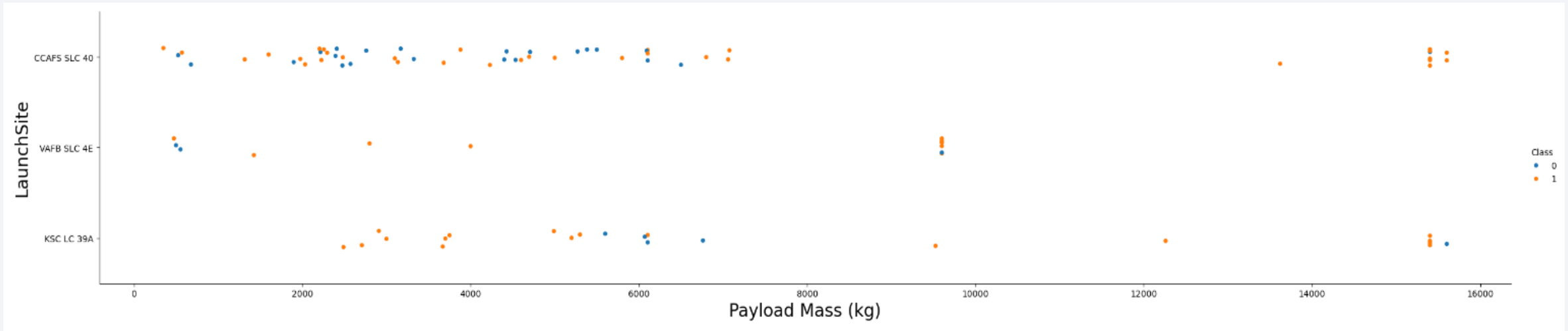
Insights drawn from EDA

Flight Number vs. Launch Site

- This scatter plot shows there is a higher success rate among Launch Sites with a greater number of flight launches



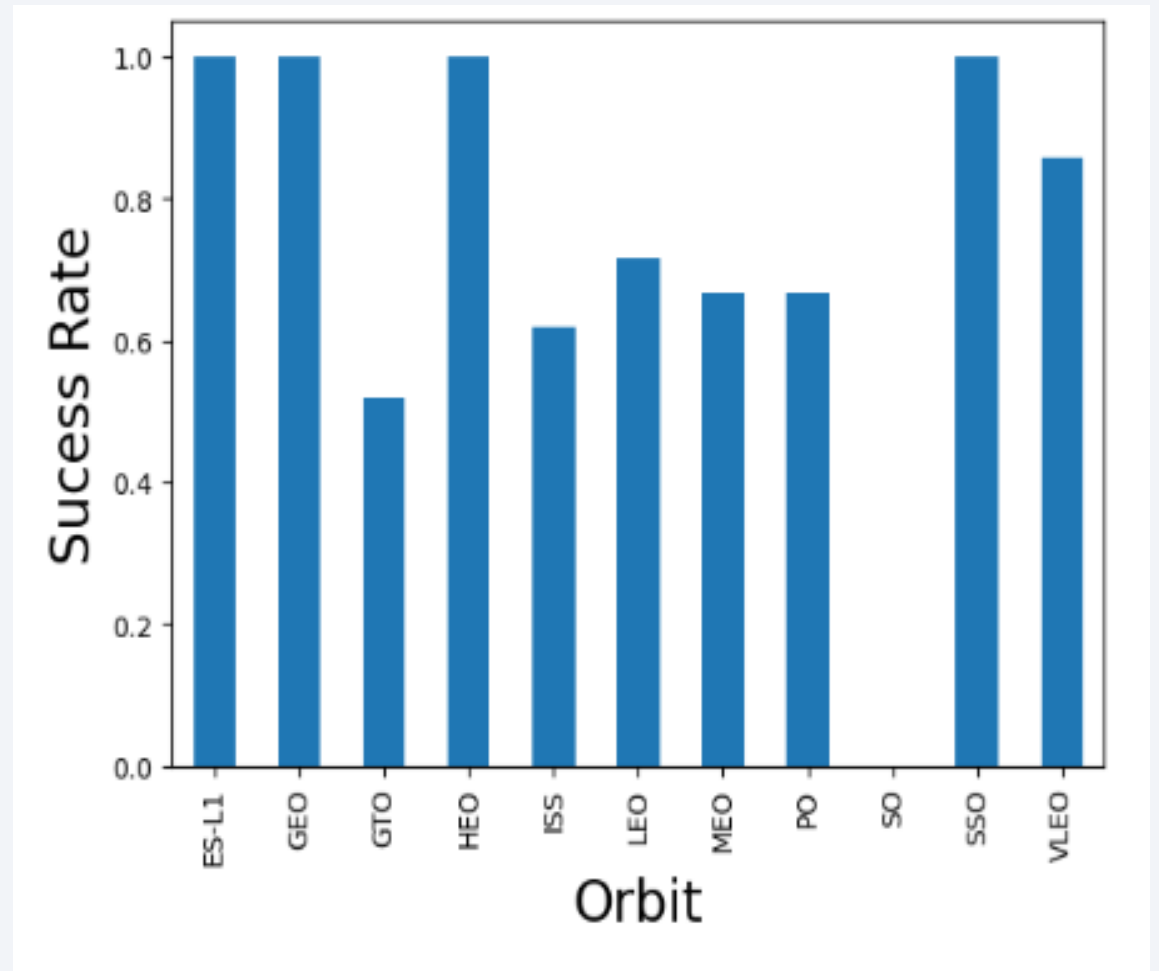
Payload vs. Launch Site



- Notably from this plot, we see that CCAFS SLC 40 has a very high success rate of flights with very heavy payloads
- At Launch Site VAFBSLC 40, no launches have been at or above 10,000 kg

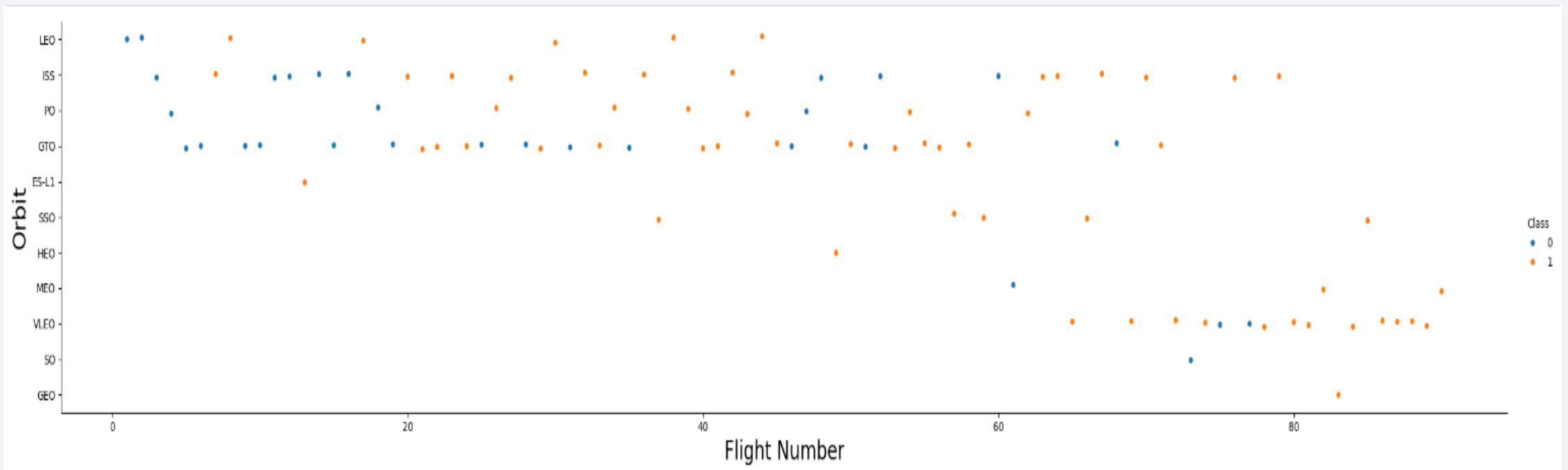
Success Rate vs. Orbit Type

- We can see the most successful orbit patterns are:
 - ES-L1
 - GEO
 - HEO
 - SSO



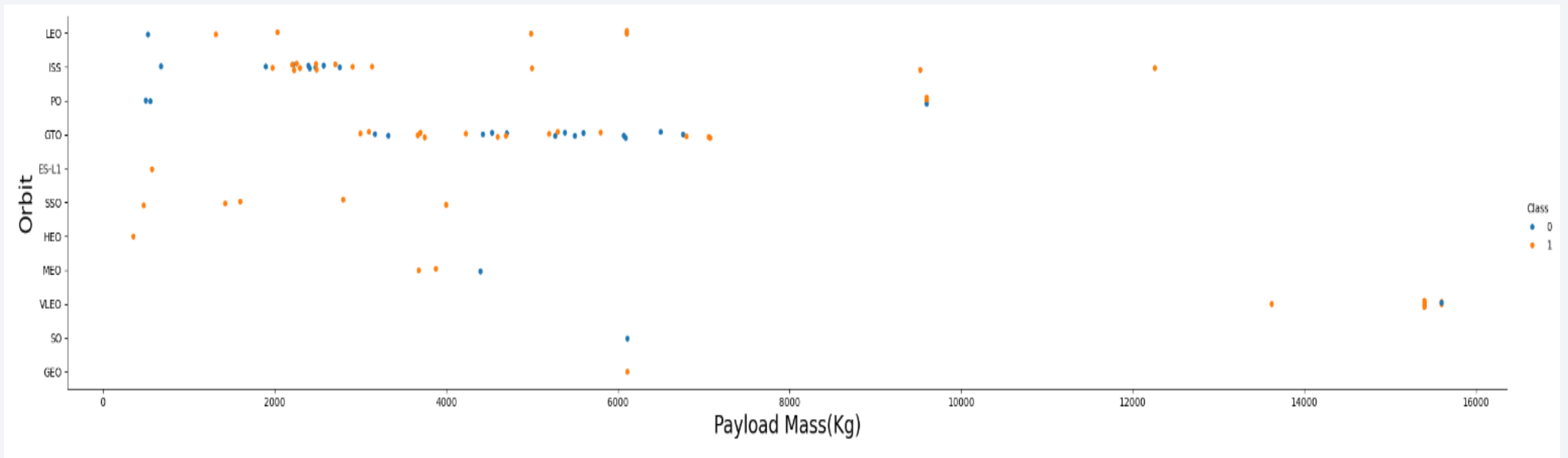
Flight Number vs. Orbit Type

- Some orbit types have a high success rate over multiple flights (LEO, VLEO) whereas other orbit types high success rate comes alongside a small sample (SSO, HEO, GEO)



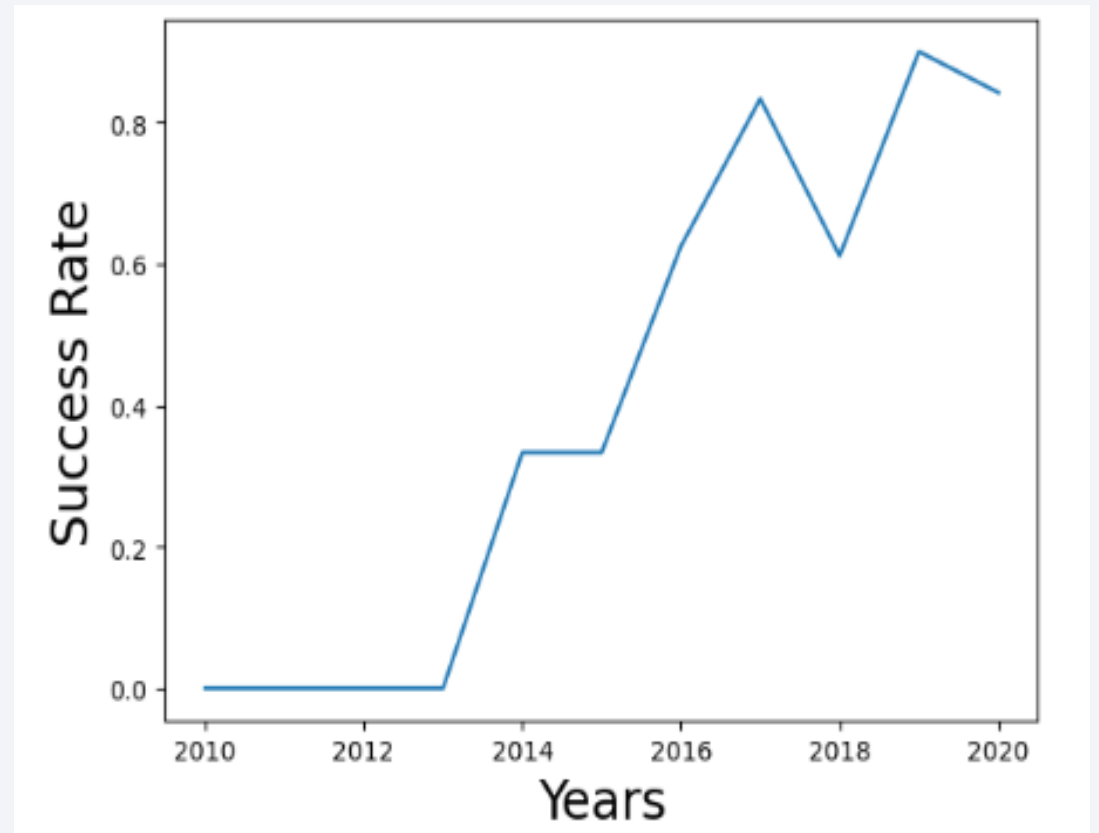
Payload vs. Orbit Type

- For heavier payloads, VLEO, ISS and PO orbits all achieve very high success rates
- Inversely, those same 3 orbit types have the lowest success rate at lighter payloads ($> 2000\text{kg}$)



Launch Success Yearly Trend

- From 2013 to 2017 there is rapid and mostly steady increase in success rates
- Success rate drops approximately 20% during 2017 but recovers the following year



All Launch Site Names

- In this query, DISTINCT was utilized to return the unique launch sites
- The commented code above was periodically used for double checking column names while writing the queries that follow

```
#%sql SELECT * FROM SPACEXTABLE LIMIT 1;  
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5
```

```
* sqlite:///my_data1.db  
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- Utilizing LIKE within the WHERE clause in the query above, we were able to filter for launch sites that begin with the letters 'CCA'
- LIMIT will limit the results to 5 rows at most

Total Payload Mass

- To calculate the total payload mass for the customer 'NASA (CRS)', the SUM function is utilized on the PAYLOAD_MASS__KG_ column, and the customer is filtered within the WHERE clause before summing the values, so we only get the values where customer = 'NASA (CRS)'.
- NASA (CRS) has a total payload mass of 45,596 kg

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) total_payload_mass FROM SPACEXTABLE WHERE customer = 'NASA (CRS)'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
total_payload_mass
```

```
45596
```

Average Payload Mass by F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) avg_payload_mass FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
avg_payload_mass
```

```
2928.4
```

- Somewhat similarly to the last query, this query filters by Booster Version and applies the AVG function to the PAYLOAD_MASS__KG_ column values after filtering
- The average payload mass of Booster Version 'F9 v1.1' is 2,928 kg

First Successful Ground Landing Date

- In this query, we filter by the Landing Outcome in the WHERE clause to 'Success (ground pad)' and apply the MIN function to the date column, retrieving the date of the first successful ground landing
- The first successful ground landing date was on December 22nd, 2015

```
%sql SELECT MIN("Date") first_landing FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
first_landing
```

```
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- Below, we filter the landing outcome for 'Success (drone ship)' and we also want to filter only for payloads between 4000 and 6000 kg
- We select the Booster Versions that meet this criteria, and added the payload mass and landing outcome columns to ensure the results are filtered properly
- There are 4 booster versions that have had successful landings of a drone ship with a payload mass between 4000 and 6000 kg

```
%sql SELECT "Booster_Version", PAYLOAD_MASS_KG_, "Landing_Outcome" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND (PAYLOAD_MASS_KG_ BETWEEN 4000 AND 6000)
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Booster_Version	PAYLOAD_MASS_KG_	Landing_Outcome
F9 FT B1022	4696	Success (drone ship)
F9 FT B1026	4600	Success (drone ship)
F9 FT B1021.2	5300	Success (drone ship)
F9 FT B1031.2	5200	Success (drone ship)

Total Number of Successful and Failure Mission Outcomes

- To count the # of successful and unsuccessful missions, we select the Mission_Outcome column as well as utilize the COUNT function to count the # of each outcome after they are grouped by each mission outcome (achieved using GROUP BY in the from clause)
- There are 100 successful mission outcomes and only 1 failure

```
%sql SELECT "Mission_Outcome", COUNT("Mission_Outcome") 'Count' FROM SPACEXTABLE GROUP BY "Mission_Outcome"
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- Utilizing a subquery, we're able to calculate the MAX payload mass, and then SELECT DISTINCT booster versions with a payload mass equal to the maximum payload weight, which is 15,600 kg

```
%sql SELECT DISTINCT "Booster_Version", PAYLOAD_MASS_KG_ FROM SPACEXTABLE WHERE PAYLOAD_MASS_KG_ = (SELECT MAX(PAYLOAD_MASS_KG_) max_payload FROM SPACEXTABLE)
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Booster_Version	PAYLOAD_MASS_KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

```
%sql SELECT SUBSTR(Date, 6, 2) month, Date, "Booster_Version", "Launch_Site", "Landing_Outcome" FROM SPACEXTABLE WHERE SUBSTR(Date, 0,5) = '2015' AND "Landing_Outcome" = 'Failure (drone ship)'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

month	Date	Booster_Version	Launch_Site	Landing_Outcome
01	2015-01-10	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	2015-04-14	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

- The SUBSTR function allowed us to pull the month out of our date column values as well as filter by just the year in our date column values
- This allows us to display the several columns of information related to landing outcomes of 'Failure (drone ship)' during 2015
- Only 2 boosters had a failure to land in 2015, one in January, one in April

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- We wanted to rank the landing outcomes between the dates 2010-06-04 and 2017-03-20 in descending order
- GROUP BY allows grouping by the landing outcome, ORDER BY allows us to order the results by the total column, which is calculated using a COUNT function in the SELECT clause
- We also add a filter in the WHERE clause for the specified date range

```
%sql SELECT "Landing_Outcome", COUNT("Landing_Outcome") total FROM SPACEXTABLE WHERE (DATE BETWEEN '2010-06-04' AND '2017-03-20') GROUP BY "Landing_Outcome" ORDER BY total DESC
```

* sqlite:///my_data1.db
Done.

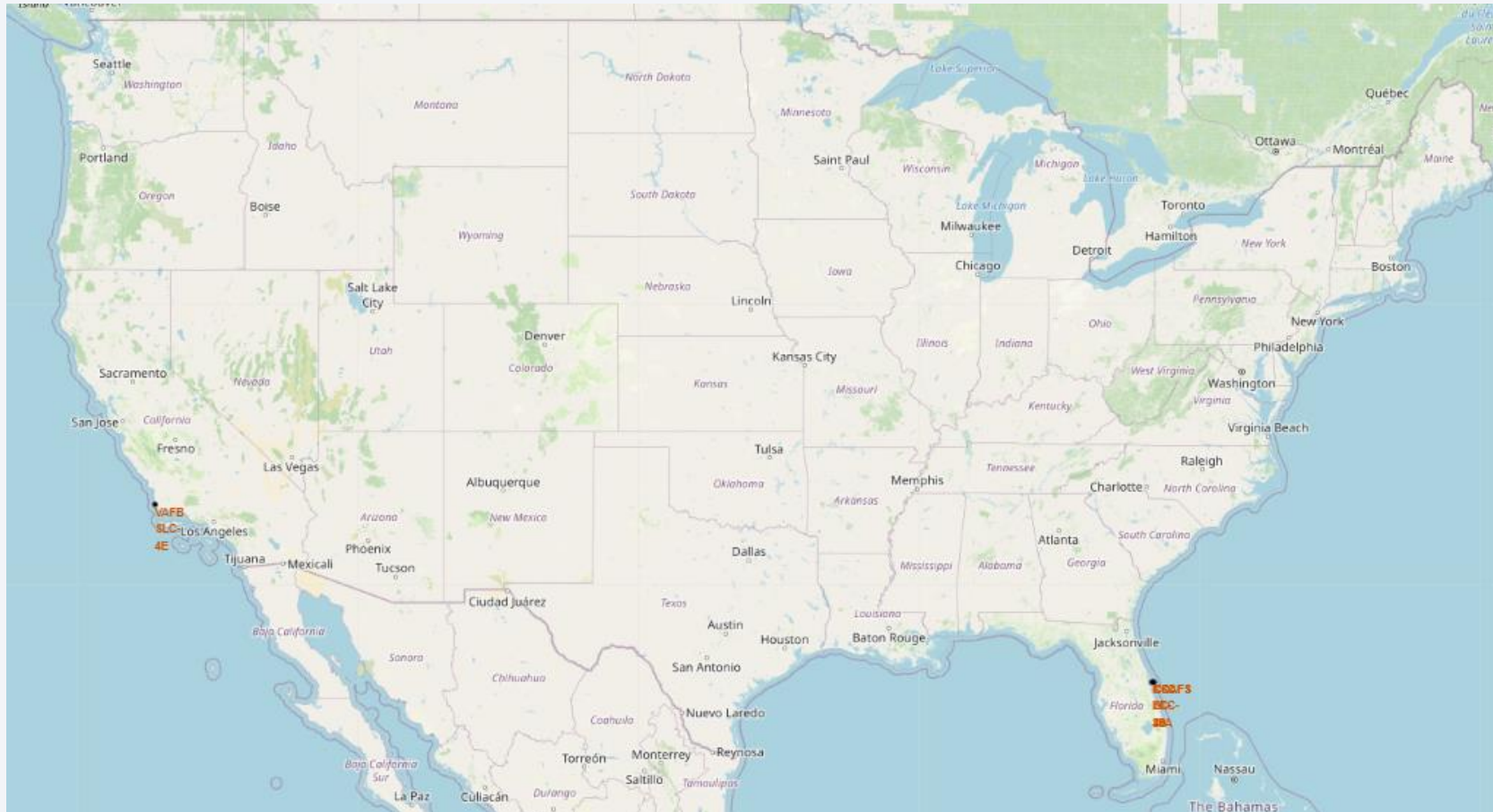
Landing_Outcome	total
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

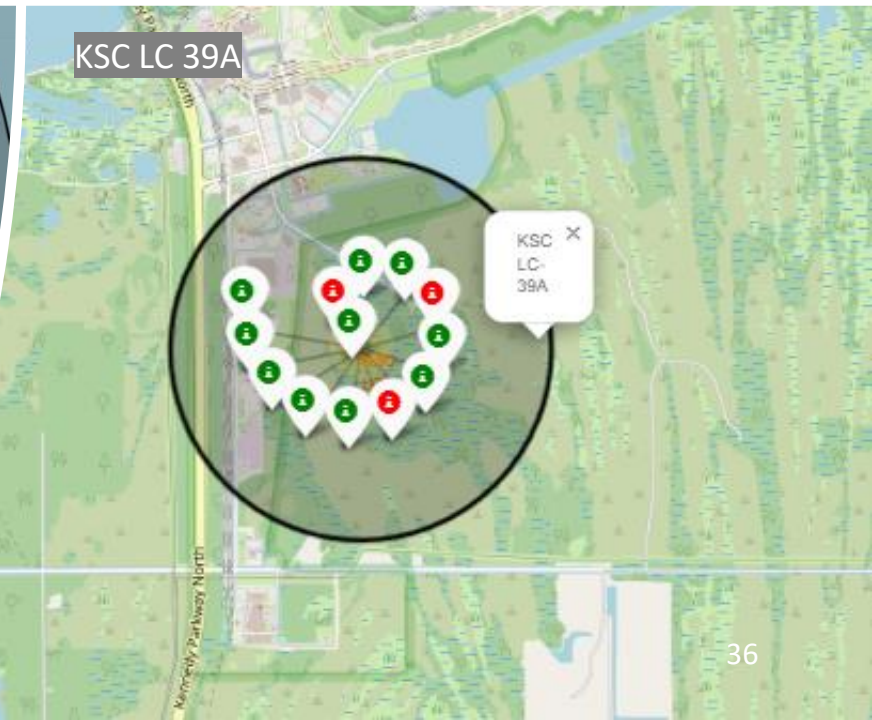
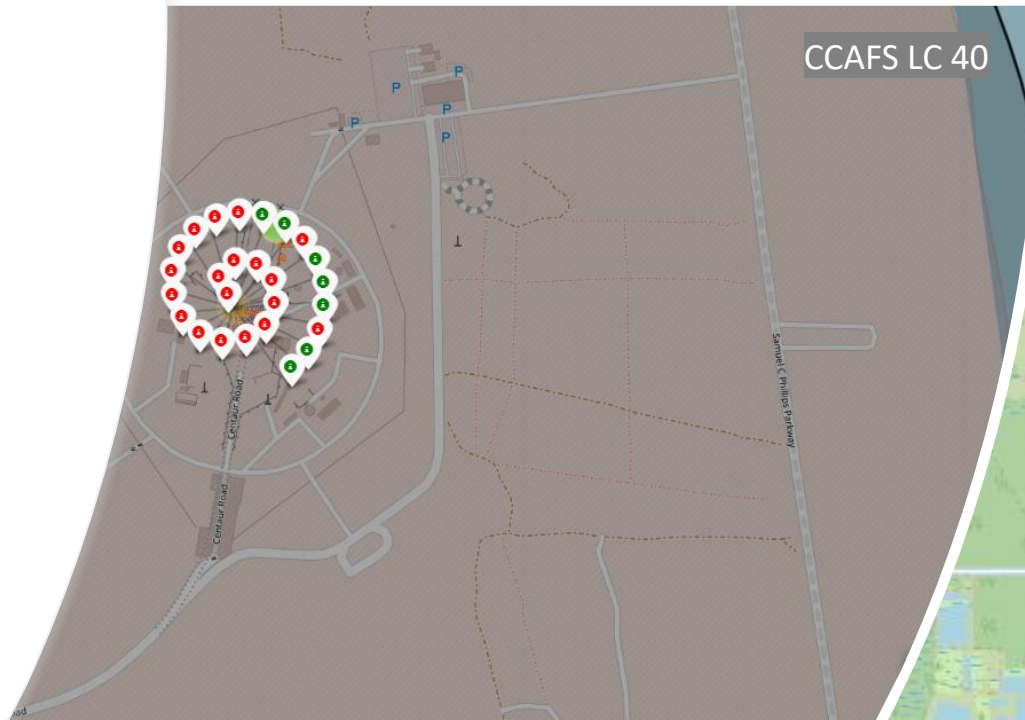
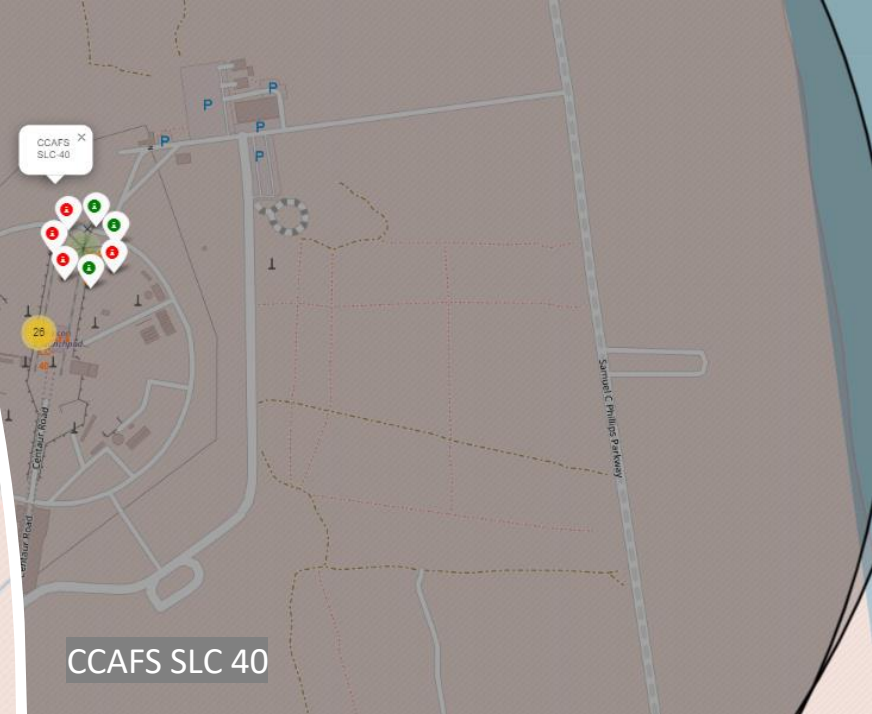
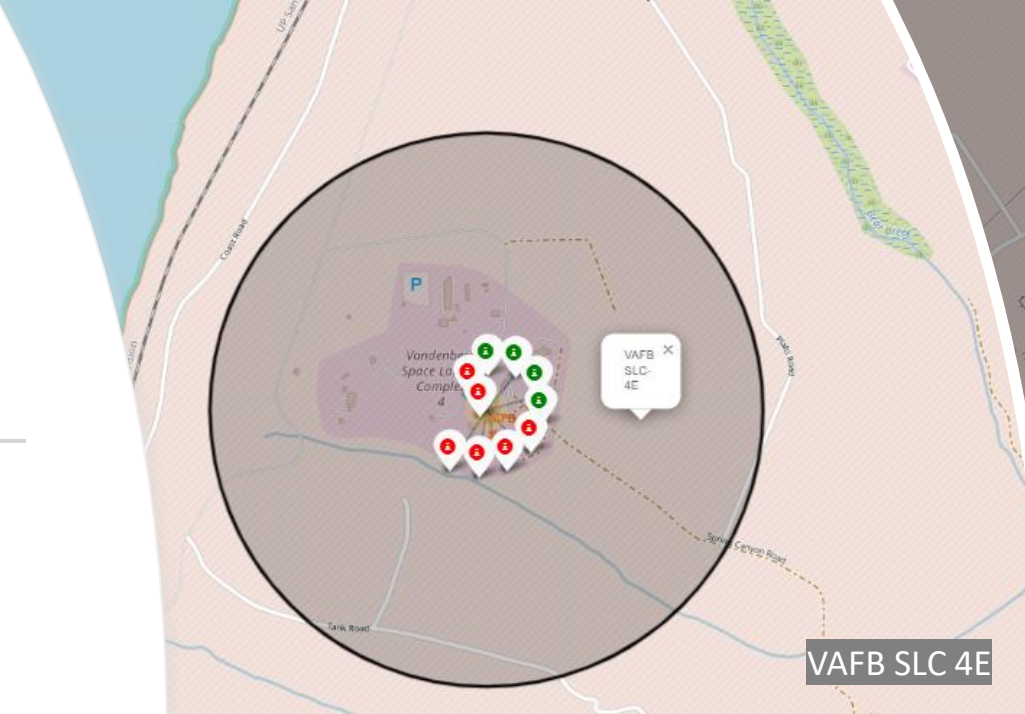
Launch Site Locations



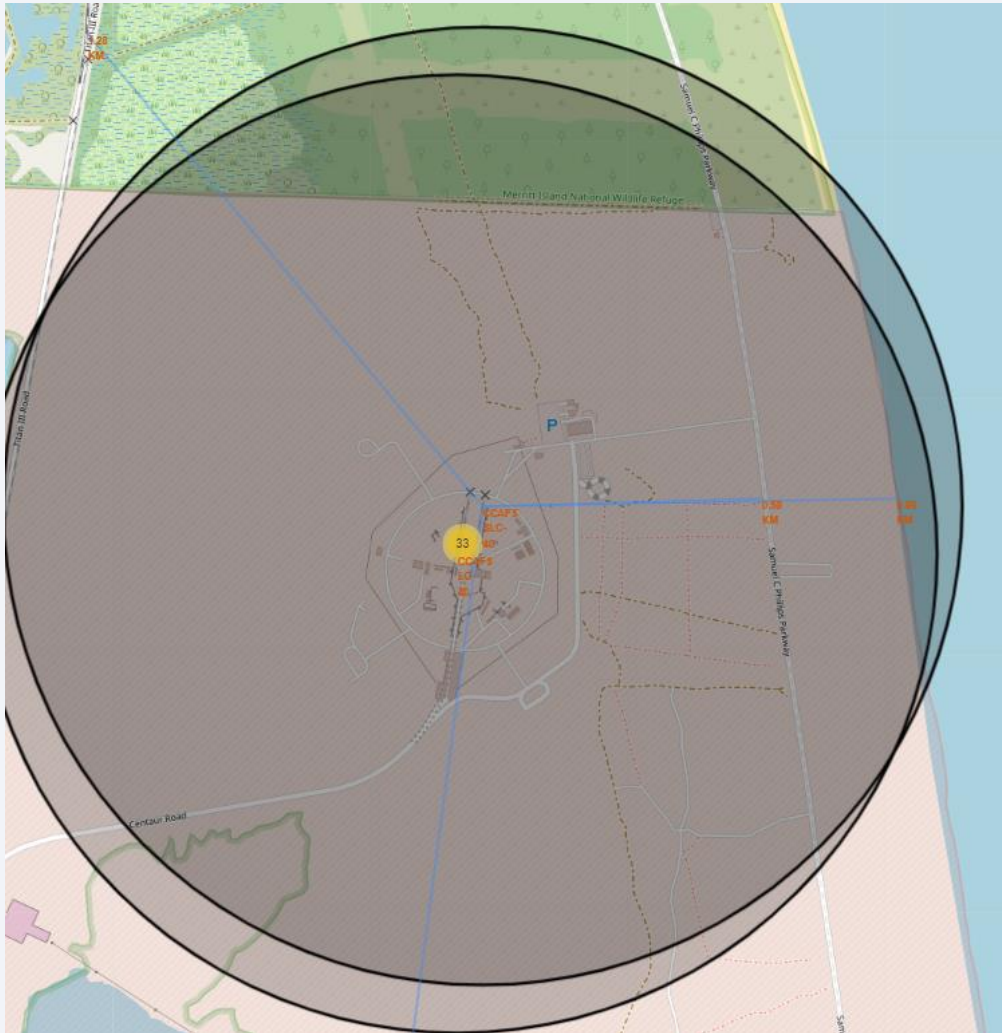
- 3 of the test sites are in Florida, 1 is in California
- All sites are in relatively close proximity to the ocean and the equator

Successes and Failures by Launch Site

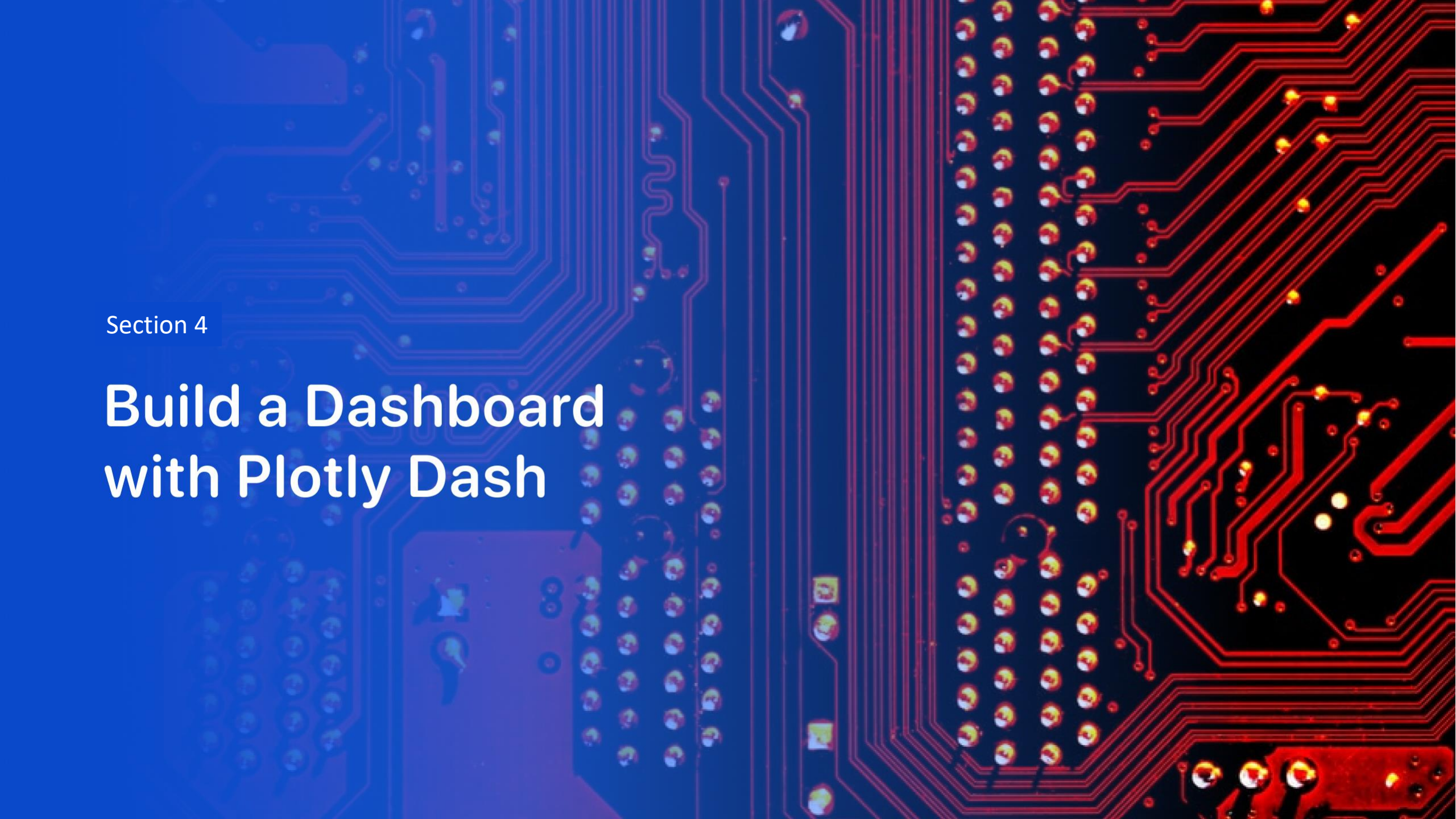
- Green markers show a successful launch, whereas red markers indicate a failed launch
- The California launch site can be seen in the bottom right image
- CCAFS SLC 40 and CCAFS LC 40 are in very close proximity to one another



What's near a Launch Site? What isn't?



- The thin blue lines on the right indicate the distance to the nearest coastline (.86km), the nearest highway (.58km), and the nearest railway (1.28km) for the two furthest east launch sites
- Notably, all 4 launch sites are very near at least one highway, one railway and are all on the coastline
- The highways and railways are ideal for bringing in heavy cargo and for workers to get to work
- All 4 launch sites are far from populous heavy areas and close to the ocean which are both likely safety precautions

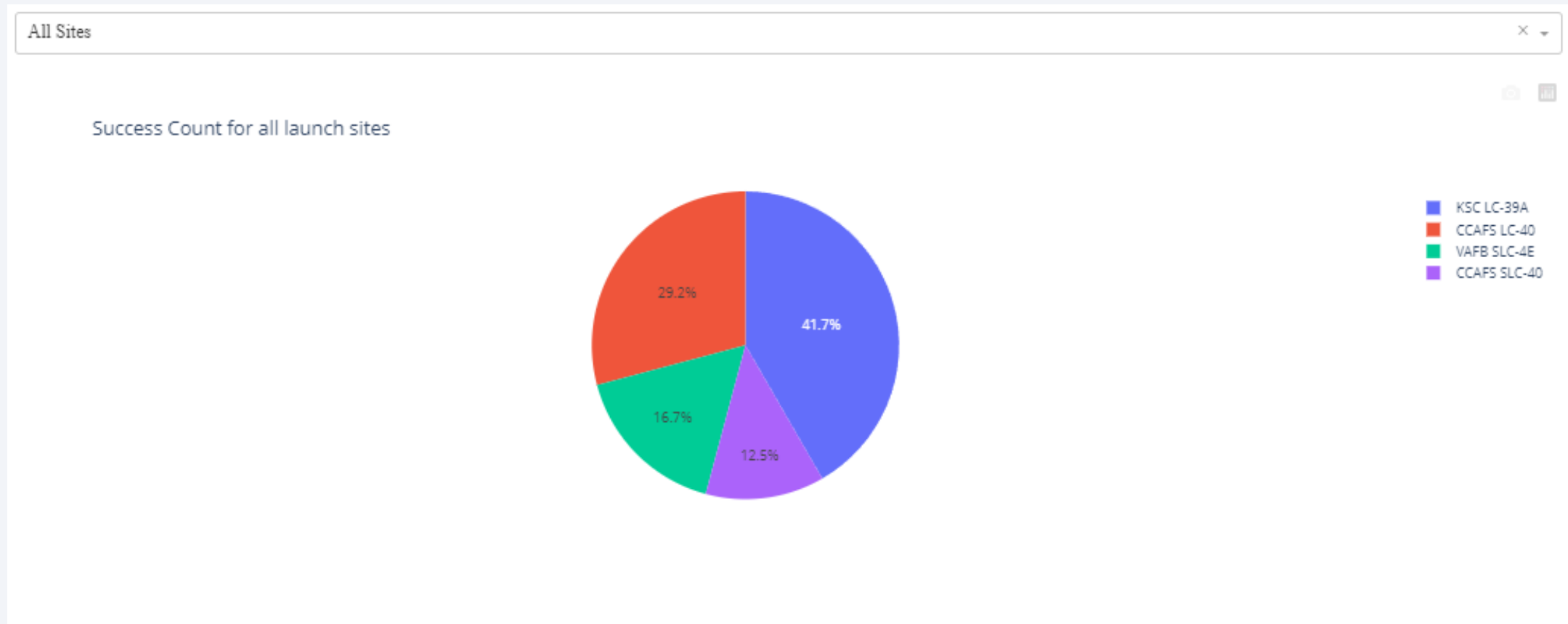
The background of the slide is a close-up, artistic photograph of a printed circuit board (PCB). The board is dark, and the intricate circuit traces are highlighted in a vibrant, glowing red. Numerous small, cylindrical components, likely capacitors or resistors, are visible, some of which also appear to be glowing. The overall aesthetic is high-tech and digital.

Section 4

Build a Dashboard with Plotly Dash

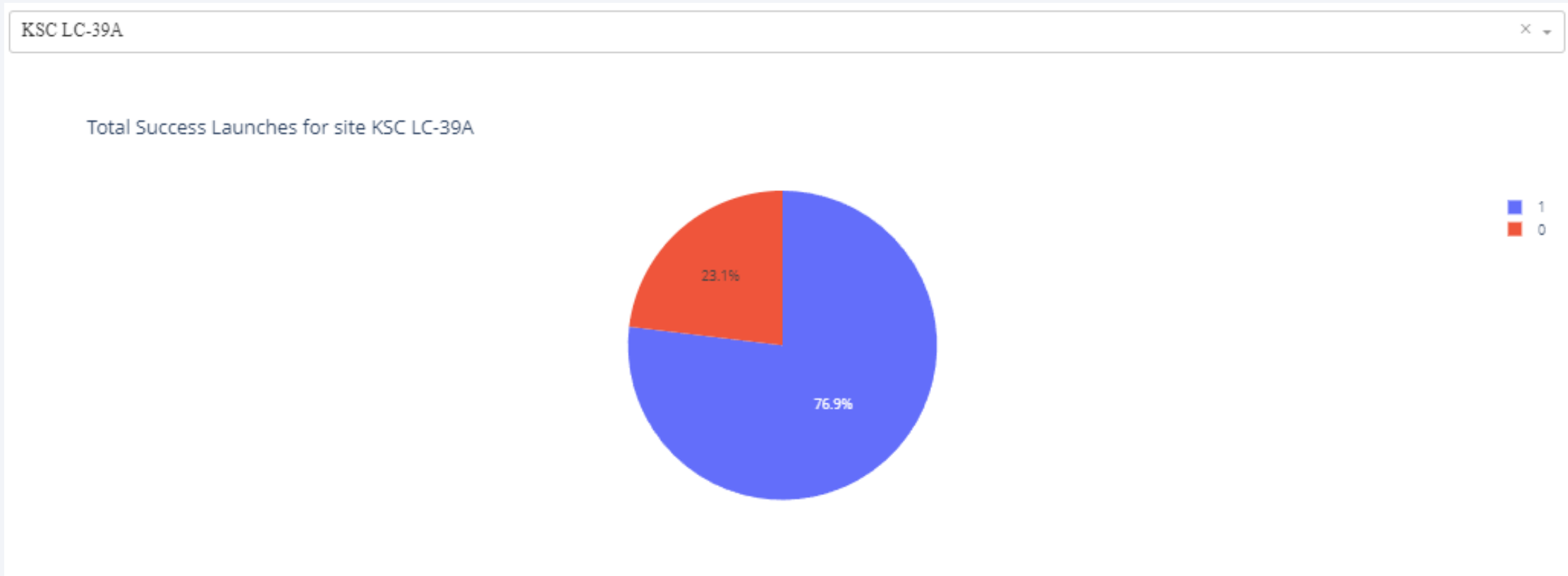
Successful Launches by Site

- The pie chart below shows that KSC LC 39 A has the highest rate of successful launches



Closer look at KSC LC 39 A

- Launch Site KSC LC 39 A has a success rate of about 77% which represents about 3 successful launches for every 1 failed launch



Payload and Launch Outcome



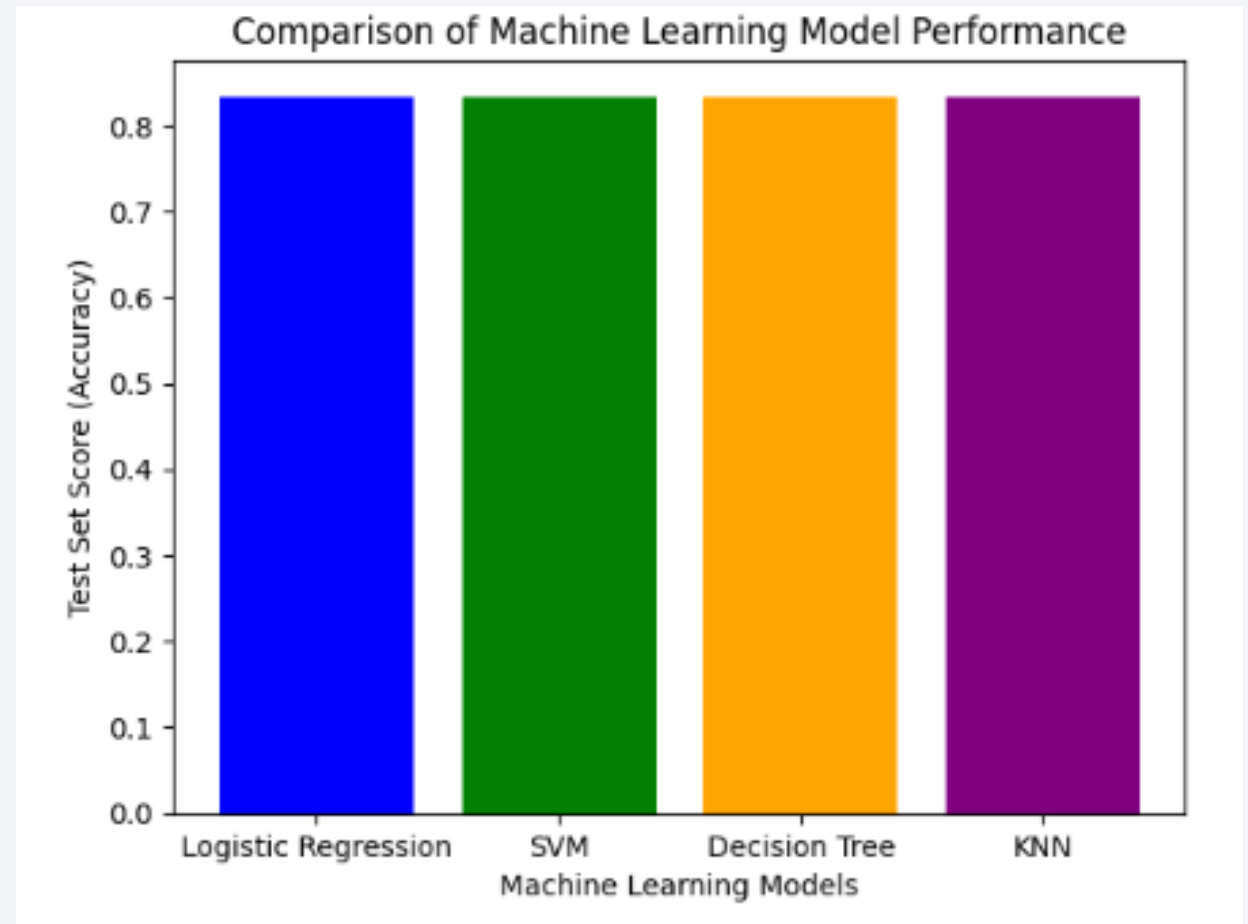
- Top Left: Payload Mass 0 – 5,000 kg
- Top Right: Payload Mass 5,000 – 10,000 kg
- Bottom Left: All Payload Masses 0 – 10,000 kg

Section 5

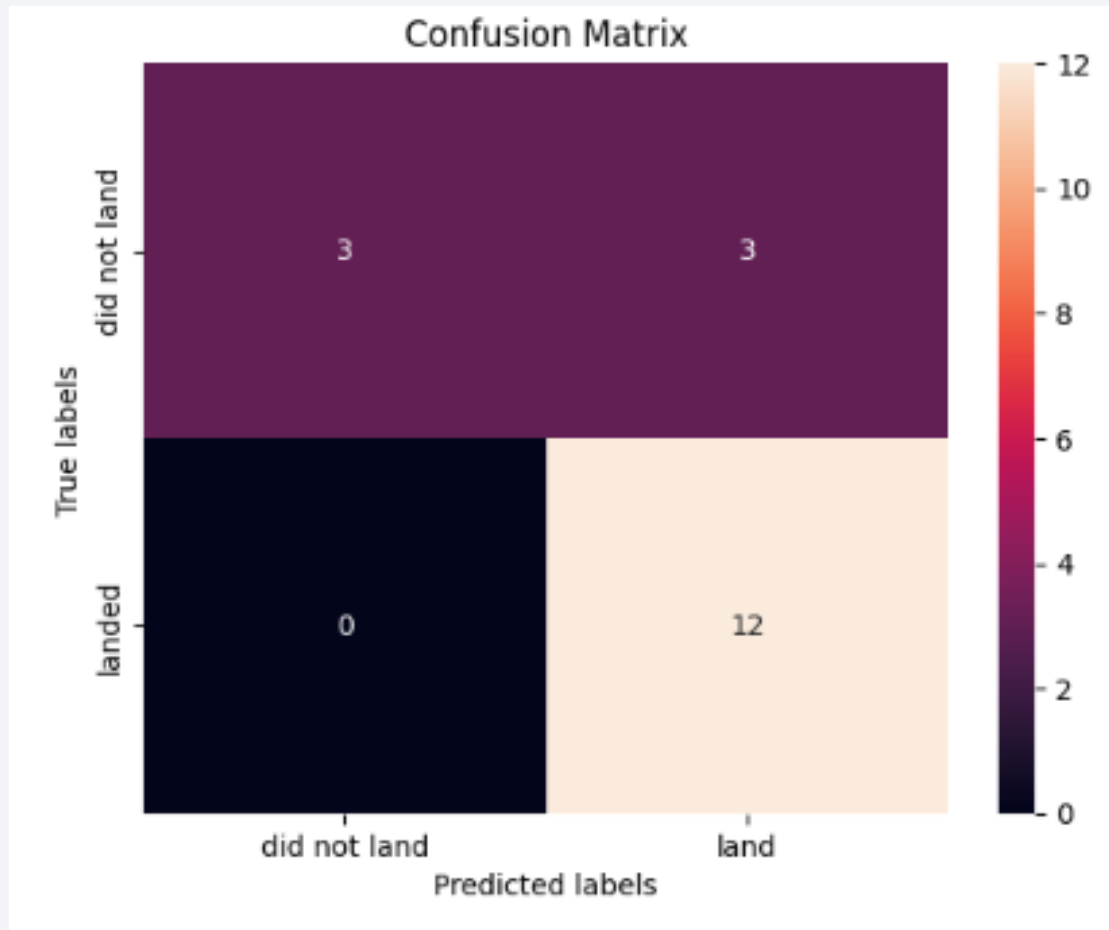
Predictive Analysis (Classification)

Classification Accuracy

- All 4 classification models in testing had the same accuracy score of approximately 83%
- No model outperforms another in this context



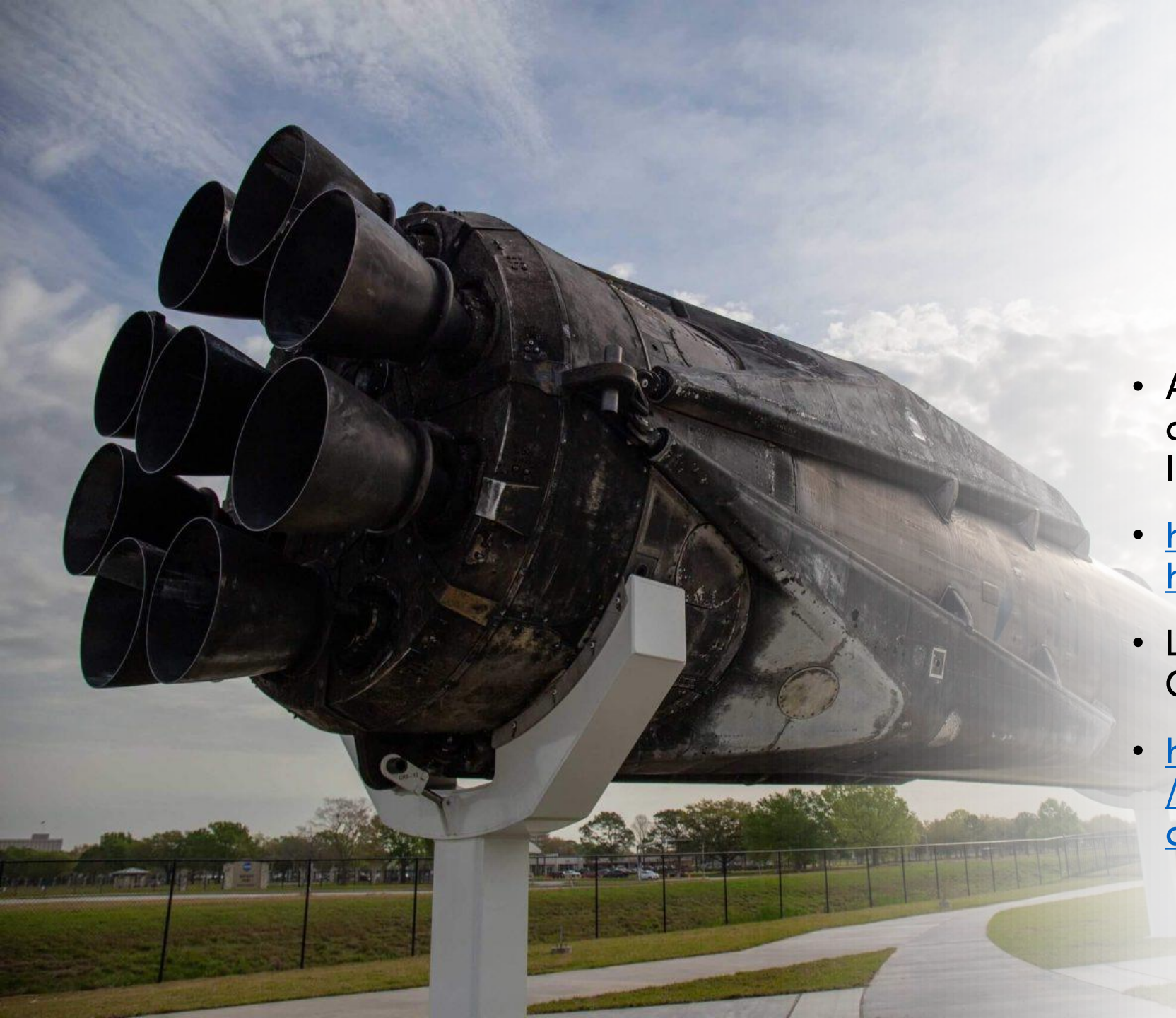
Confusion Matrix



- As all models had the same accuracy score, they also had the same confusion matrix in all 4 cases
- The classifiers give a high number of false positives in each case

Conclusions

- Launch Sites with the most launches tend to have the highest success rate
- LEO, PO, and ISS orbits have the highest success rates for heavier payloads
- Launch Sites are always distanced from high population areas and are always positioned near a coastline for easy retrieval of rockets and safety reasons
- Launch Site KDC LC 39 a has the highest success rate of any site
- All classification models yielded a similar level of accuracy, so any one of them is appropriate for predicting if SpaceX will be able to reuse the first stage on a Falcon 9 rocket launch or not



Appendix

- All files related to this project can be found at the following link:
- <https://github.com/jtnickel/Python/tree/main>
- Link to Capstone Project on Coursera:
- <https://www.coursera.org/learn/applied-data-science-capstone/>

Thank you!

